

$f : G \rightarrow \text{Sym}(A)$ s.t.

$$f(g) = \sigma_g$$

where $\sigma_g(a) = g * a = ga$

Let $f(g_1) = f(g_2)$

$$\Rightarrow \sigma_{g_1} = \sigma_{g_2}$$

$$\Rightarrow \sigma_{g_1}(a) = \sigma_{g_2}(a)$$

$$\Rightarrow g_1 * a = g_2 * a$$

$$\Rightarrow g_1 * e = g_2 * e \quad (\text{put } a = e \in A)$$

$$\Rightarrow g_1 e = g_2 e$$

$$\Rightarrow g_1 = g_2$$

$\therefore f$ is one-one

$$\Rightarrow \ker f = \{e\}$$

Also, by fundamental theorem of group homomorphism

$$G / \ker f \cong f(G)$$

$$G / \{e\} \cong f(G)$$

$$G \cong f(G)$$

So, $G \cong f(G) \leq \text{Sym}(A) = \text{Sym}(G)$.

Kernel of an action

Let $* : G \times A \rightarrow A$ be a group action, then Kernel of $*$ is defined as

$$\ker * = \{g \in G \mid g * a = a, \forall a \in A\}$$

i.e. those elements of G that fix all element of A .

$$\ker * \leq G$$

Proof: $e \in \ker * \implies \ker * \neq \phi$

Let $g_1, g_2 \in \ker *$. Then

$$g_1 * a = a \quad \text{and} \quad g_2 * a = a, \quad \forall a \in A$$

Now,

$$\begin{aligned} (g_1 g_2) * a &= g_1 * (g_2 * a) \\ &= g_1 * a \\ &= a \quad \forall a \in A \end{aligned}$$

$$\implies g_1 g_2 \in \ker *$$

$$\text{Also, } g_1 * a = a$$

$$\begin{aligned} \implies g_1^{-1} * (g_1 * a) &= g_1^{-1} * a \\ \implies (g_1^{-1} g_1) * a &= g_1^{-1} * a \\ \implies e * a &= g_1^{-1} * a \\ \implies a &= g_1^{-1} * a \\ \implies g_1^{-1} &\in \ker * \end{aligned}$$

$$\therefore \ker * \leq G.$$

Theorem: If f is the corresponding permutation representation of $*$ then

$$\ker f = \ker *$$

Proof:

$$f : G \rightarrow \text{Sym}(A)$$

$$f(g) = \sigma_g$$

$$\sigma_g(a) = g * a$$

$$\begin{aligned} \ker f &= \{g \in G \mid f(g) = I_A\} \\ &= \{g \in G \mid \sigma_g = I_A\} \\ &= \{g \in G \mid \sigma_g(a) = I_A(a), \forall a \in A\} \\ &= \{g \in G \mid g * a = a, \forall a \in A\} \end{aligned}$$

So,

$$\ker f = \ker *$$

Orbit and Stabilizers

Def: Let $*$ be a group action of a group G on a set A and let $a \in A$. Then the set

$$G_a = \{g \in G \mid g * a = a\}$$

is called **stabilizer** of a in G .

Theorem: $G_a \leq G$.

Proof:

Since $e * a = a$ for any $a \in A$

$$\therefore e \in G_a \implies G_a \neq \phi$$

Let $g_1, g_2 \in G_a$ for any $a \in A$
 $\Rightarrow g_1 * a = a$ and $g_2 * a = a$
Now,

$$\begin{aligned}(g_1 g_2) * a &= g_1 * (g_2 * a) \quad (\text{as } * \text{ is group action}) \\ &= g_1 * a \\ &= a\end{aligned}\tag{1}$$

$\therefore g_1 g_2 \in G_a$
Also, $g_1 * a = a$

$$\begin{aligned}\Rightarrow g_1^{-1} * (g_1 * a) &= g_1^{-1} * a \\ \Rightarrow (g_1^{-1} g_1) * a &= g_1^{-1} * a \\ \Rightarrow e * a &= g_1^{-1} * a \\ \Rightarrow g_1^{-1} * a &= a\end{aligned}\tag{2}$$

$\therefore g_1^{-1} \in G_a$
From (1) & (2),

$$G_a \leq G$$

Example: If G acts on itself by conjugation i.e.

$$g * a = gag^{-1}, \quad \forall g \in G, a \in A (= G)$$

Then for any $a \in A$

$$\begin{aligned} G_a &= \{g \in G \mid g * a = a\} \\ &= \{g \in G \mid gag^{-1} = a\} \\ &= \{g \in G \mid ga = ag\} \end{aligned}$$

$G_a = N(a)$ (Normalizer of a)

Def: Let G be a group acting on a set A under $*$. Then for any $a \in A$, the set

$$\begin{aligned} G \cdot a &= \{x \in A \mid g * a = x, \text{ for some } g \in G\} \\ &= \{g * a \mid g \in G\} \end{aligned}$$

is known as **orbit** of a in G or orbit of G containing a .

Theorem (Orbit Stabilizer theorem): Let G be a group that acts on a set A . Let $a \in A$, then there exists a one-one onto map from $G \cdot a$ to the set of all left cosets of G_a in G .

Proof:

$$G \cdot a = \{g * a \mid g \in G\}$$

$$G_a = \{g \in G \mid g * a = a\}$$

$$G_a \leq G$$

So, G/G_a = set of all left cosets of G_a in G

Define: $\phi : G \cdot a \rightarrow G/G_a$ as

$$\phi(g * a) = gG_a, \quad g \in G$$

Let $g_1 * a = g_2 * a$

$$\Rightarrow g_2^{-1} * (g_1 * a) = g_2^{-1} * (g_2 * a)$$

$$\Rightarrow (g_2^{-1}g_1) * a = e * a = a$$

$$\Rightarrow g_2^{-1}g_1 \in G_a$$

$$\Rightarrow g_1G_a = g_2G_a$$

$$\Rightarrow \phi(g_1 * a) = \phi(g_2 * a)$$

$\therefore \phi$ is a well-defined map.

Now,

$$\begin{aligned}\phi(g_1 * a) = \phi(g_2 * a) &\Rightarrow g_1 G_a = g_2 G_a \\ &\Rightarrow g_2^{-1} g_1 \in G_a \\ &\Rightarrow (g_2^{-1} g_1) * a = a \\ &\Rightarrow g_2 * ((g_2^{-1} g_1) * a) = g_2 * a \\ &\Rightarrow (g_2 g_2^{-1} g_1) * a = g_2 * a \\ &\Rightarrow g_1 * a = g_2 * a\end{aligned}$$

$\therefore \phi$ is one-one.

Let $t \in G/G_a$

$$\begin{aligned}\Rightarrow t &= gG_a \text{ for some } g \in G \\ \Rightarrow g * a &\in G \cdot a \text{ s.t. } \phi(g * a) = gG_a = t\end{aligned}$$

$\therefore \phi$ is onto.

Note: If G is a finite group then

$$|G \cdot a| = [G : G_a] = \frac{|G|}{|G_a|}$$

$$\Rightarrow \boxed{|G| = |G \cdot a| \cdot |G_a|}$$